

# BandhakaviSaiAshish

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## SKILLS

|                     |                                                                         |
|---------------------|-------------------------------------------------------------------------|
| <b>Languages:</b>   | Python                                                                  |
| <b>Frameworks:</b>  | TensorFlow, scikit-learn, Gradient Boost                                |
| <b>Soft Skills:</b> | Problem-Solving Skills, Critical Thinking, Adaptability                 |
| <b>Software:</b>    | Fusion360, Auto Cad, Automation-anywhere, Ui-Path, Canvas, PCB Designer |

## INTERNSHIP

### Fundamentals of Python In Machine Learning

April- June 2024

#### Project

**About:** Developed and contributed to a project focused on making generative AI accessible to everyone. The project aimed to create easy-to-use applications and interfaces for generative models, enabling users to generate creative content like text, images, and other media. Worked on enhancing the user experience and optimizing AI models for various practical applications.

- **Key Responsibilities:**
  - Worked on training and fine-tuning generative models to produce high-quality outputs.
  - Developed intuitive user interfaces for AI-driven content generation tools.
  - Collaborated with the team to optimize model performance and user interaction.
  - Conducted testing and evaluation to ensure model outputs met quality standards.
- **Tech stacks used:** Python, TensorFlow, PyTorch

## PROJECTS

### Mood-Scape-Predictive Modeling of Depression and Anxiety via Digital Biomarkers

September – Present 2025

Developed a machine learning and deep learning-based predictive system to identify early signs of depression and anxiety using digital biomarkers such as voice tone, facial expressions, and behavioral patterns. The model aims to assist in mental health screening and preventive interventions.

Mood Disorders, which include depression and anxiety, are the most frequent and disabling health issues all around the world and have significantly reduced the quality of life of the individuals affected.

Traditional diagnosis and treatment of the disorders are almost entirely based on questionnaires, clinical interviews, and self-reporting and are subjective and not very specific.

Moreover, due to the complexity of the biological, behavioral, and social processes of mental health, on-time and accurate diagnosis is still a great challenge of healthcare.

Its ultimate vision is the delivery of scalable, ubiquitous, and efficient instruments capable of augmenting existing clinical practices, boosting treatment adherences, and augmenting individuals with mood disorders' quality of life.

### Drug Screening using Stem Cell Technology

January– May 2025

Stem cell technology has revolutionized drug screening by providing more accurate and human-relevant models for testing drug efficacy and toxicity. Unlike traditional methods that rely heavily on animal models, stem cell-based approaches utilize human cells, offering insights that are more predictive of human responses.

**Disease Modeling:** By differentiating stem cells into specific cell types affected by diseases, researchers can create in vitro models that closely mimic human pathologies. For instance, motor neurons derived from patient-specific stem cells have been used to study motor neuron disease (MND), enabling rapid and accurate testing of potential treatments.

**Toxicity Testing:** Stem cell-derived cardiomyocytes (heart cells) and hepatocytes (liver cells) are employed to assess the cardiotoxicity and hepatotoxicity of new drug candidates. This approach helps in identifying adverse effects early in the drug development process, potentially reducing the high failure rates in later stages.

**High-Throughput Screening:** The scalability of stem cell cultures allows for high-throughput screening of numerous compounds simultaneously. This efficiency accelerates the identification of promising drug candidates and facilitates the study of various diseases.

**Personalized Medicine:** Induced pluripotent stem cells (iPSCs) can be generated from individual patients, providing a platform for testing how specific genetic backgrounds respond to different drugs. This personalized approach enhances the development of tailored therapies and reduces the risk of adverse drug reactions.

While stem cell-based drug screening offers significant advantages, challenges remain. Ensuring the consistent differentiation of stem cells into mature, functional cell types is complex. Additionally, ethical considerations, particularly concerning the use of embryonic stem cells, necessitate careful oversight and adherence to regulatory guidelines.

### Building robust machine translation and document search systems: Combining neural networks and vector space models

August-December 2024

A general trend for such systems was the exploitation of new techniques in this area of machine learning, specifically neural networks, which hopefully would advance traditional VSMs to their application much better for tasks like document search. Indeed, neural networks, especially in semantic search, work with dense vector embeddings representing documents and queries in a space in which semantically related items are closer together. This is much better compared to previous keyword-based search models since it returned so many applicable results where an actual word could not be found.

Practically, OpenSearch and vector databases fasten similarity searches within large spaces of high dimension and implement these methods. Applications such as search into the document best fit using these approaches since information on semantics and context improves the user experience substantially.

### Recommendation System that Suggests Recipes

August-December 2024

Recommendation systems utilize user profiles and filtering technology to support users in locating suitable information from an extensive amount of data. User profile is of great importance in the recommendations. This paper employs two methods to recommend recipes whereby user preferences are introduced as ratings for comparison so that it can be established as to which one will better suit the data homogeneity.

Two methods-the item-based method and the user-based method-are used to recommend recipes. In the item-based method, we shall use the Tanimoto coefficient similarity and log-likelihood similarity to calculate the similarities between different recipes. In the user-based method, we will use Euclidean distance and Pearson correlation. Unlike earlier, user-based similarity techniques are introduced, whereby a fixed-size neighborhood and threshold-based neighborhood have been introduced. For this dataset, it was found that user-based performances exceed item based recommendations. All recipe dataset beat the simulated dataset since interaction and exchanges.

### A Review on Assessment on Knowledge, Attitude and Practice on Mobile Addiction Leading to Psychological Problems Among College Going Students

August-December 2023

Mobile addiction has become a significant concern among college students, leading to various psychological problems such as anxiety, depression, sleep disturbances, and reduced academic performance. The excessive use of smartphones impacts students' social behavior, cognitive abilities, and overall well-being. Assessing the knowledge, attitude, and practices (KAP) of students regarding mobile addiction is crucial to understanding the extent of the problem and designing effective interventions. With the advancement of technology, machine learning (ML) is increasingly being used to analyze behavioral patterns and predict addiction tendencies. ML models can help identify high-risk individuals, analyze their usage patterns, and suggest personalized interventions. By leveraging ML frameworks, researchers can develop predictive models that assess the psychological effects of mobile addiction and provide data-driven insights for mitigation strategies.

### Predicting Company Turnover using HR Data

August-December 2023

HR data to contribute to the business's bottom line, with objectives including boosting productivity, improving performance, and enhancing employee experience. I have developed expertise in using HR data to predict company turnover. By analyzing key metrics such as employee engagement scores, performance evaluations, turnover rates, and exit interview data, I have successfully identified trends and patterns that indicate high-risk individuals or departments. This predictive analysis has allowed me to proactively recommend interventions and retention strategies to reduce turnover and improve overall employee satisfaction. I am confident in my ability to leverage HR data to provide actionable insights that drive business results and effectively mitigate turnover risk.

### 3D Printer

January– May 2022

This skill shows your ability to create molds efficiently and effectively, and your understanding of different materials and machinery. Utilize expertise in operating and maintaining 3D printers to produce high-quality, precise three-dimensional objects. Proficient in interpreting digital designs and ensuring optimal performance of 3D printing processes  
Tech: Python, Blender, Fusion 360, AutoCad, 3D Printing Applications.

## CERTIFICATES

- Drug Screening Using Stem Cell Technology, IEEE, 2025
- Industrial Internet of Things 4.0, NPTEL, 2025
- Agile Project Management, University of Colorado Boulder, 2025
- Introduction to Internet of Things, NPTEL, 2024
- Cybersec Symposium Quick Heal, CompTia, April 2024
- Diabetic Retinopathy Detection with Artificial Intelligence, 2024
- Natural Language Processing with Classification and Vector Spaces, 2024
- Natural Language Processing with Probabilistic Models, 2024
- Natural Language Processing with Sequence Models, 2024
- Natural Language Processing with Attention Models, 2024
- [Analysis of Algorithms](#), 2024
- Supervised Machine Learning: Regression, 2024.
- [Static Routing Configuration Using Packet Tracer](#), 2023

## ACHIEVEMENTS

- Long Jump, Sri Vamshidhar High School, 2015, 2016, 2017, 2018 (Gold medal)
- Drug Screening Using Stem Cell Technology, IEEE, 2025

## EDUCATION

### Lovely Professional University

Bachelor of Technology in Lateral Entry- Computer Science and Engineering; CGPA: **4.00%**

Punjab, India

Since August 2023

### Nettur Technical Training Foundation

Diploma in Mechatronics; **Percentage: 67.6%**

Bangalore, Karnataka

April 2019 - March 2022

### Sri Vamshidhar High School

Matriculation; **Percentage: 67.6%**

Hyderabad, Telangana

April 2015 - March 2019